not exactly be 3 dimensionally printed, it paved the way for several other developments in the use of 3D printers in medicine.

In 2002, scientists at the Wake Forest Institute for Regenerative Medicine successfully printed a working kidney. The kidney when implanted in test animals actually filtered blood and produced diluted urine. In less than ten years, scientists have managed to print other parts of the body too. Skin, ears, bones, noses and even blood vessels have been printed by scientists across the world through 3D bio-printers that use a gel like substance made up of human cells extracted from the patient and grown in cell cultures. These cells are mixed with hydro-gel and used to print complete organs layer by layer thus spawning a whole new Bio-printing industry.

While there may be huge potential in this industry and could potentially save millions of lives and eliminate the long waiting lists one must endure to find a match, concerns have been raised about the who will regulate and set a standard of quality. Others claim that bio-printing takes the topic of playing god to whole new level.

The ability to print parts of the body using cells from the patient's body means that the chances of rejection are significantly reduced. This has lead to 3D printing being used in reconstructive surgery as well. In the case of Eric Moger of the UK who had lost almost the entire left side of his face to cancer, doctors were able to develop a nylon skull that served as a prosthesis.

Embraced by medical technology

The same is the story of an 83 year old woman who had a new jaw bone printed out of a titanium powder. The jaw bone was given a bio-ceramic coating and has all the grooves and indentations required for the muscles to reattach themselves to the prosthetic bone.

Breast cancer survivors with partial mastectomies too have benefited from 3D bio-printing as scientists at the University of Texas at El Paso have managed to use the patient's own fat cells to create a custom fitted breast implant.

Post the medical revolution that the 3D printer kicked off, open source 3D printers made their first appearing. The brainchild of Dr. Adrian Bower